

Berfin Celik

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Computer Engineer with 2+ years of professional backend experience in .NET Core and cloud-native systems, now focused on ML/AI engineering following an MSc in Computer Science & Artificial Intelligence (Distinction). Experienced in building and deploying production-grade NLP pipelines, LLM fine-tuning, RAG systems, and speech processing models.

EDUCATION

University of Bradford - MSc Computer Science & Artificial Intelligence | **Distinction** 2024 - 2025

Dissertation: Designed and evaluated an end-to-end NLP pipeline for multidialectal language understanding in low-resource languages (Turkish, Kurmanji, Zazaki), combining ASR, topic modelling, and knowledge graph construction. Fine-tuned Whisper and Wav2Vec2 on a custom corpus; implemented BERTopic, KeyBERT/YAKE, and NetworkX for semantic knowledge graphs. Evaluated with WER, CER, BLEU, and topic coherence, with human-in-the-loop validation by native speakers. ([PDF](#))

Bahcesehir University - BSc Computer Engineering 2016 - 2021

EXPERIENCE

Junior .NET Developer - Path Product & Software House Istanbul, Turkey Dec 2023 - May 2024

- Optimised critical back-end services in C# and .NET Core for high-load applications, improving response times and efficiency through performance tuning and algorithmic improvements.
- Architected a microservices structure based on Onion Architecture, improving modularity and maintainability; integrated Redis distributed caching to reduce database load and improve retrieval speeds.
- Containerised applications with Docker and collaborated within Agile/Scrum teams, enhancing deployment consistency and overall system stability.

Software Engineer - Argevi Software & Automation Istanbul, Turkey Mar 2022 - Jun 2023

- Optimised back-end systems and database schemas using advanced query tuning and algorithms, improving performance on high-traffic applications.
- Engineered and deployed a Project Management Tool (ASP.NET Core, React) and an E-commerce Platform (.NET Core, Angular) with secure payment gateways, Redis caching, and Docker containerisation.
- Implemented Kubernetes for orchestration and RabbitMQ for message brokering; gained exposure to OPC and TCP/IP protocols within an industrial automation context.

IT Intern - Turkuvaz Media Group Istanbul, Turkey Dec 2020 - Mar 2021

- Performed rigorous user testing on updated application components and monitored database performance, contributing to the release of stable, high-quality digital media products.
- Assisted in optimising MySQL queries, improving data retrieval speeds and overall system efficiency.

Student Assistant - Bahcesehir University Engineering Faculty Istanbul, Turkey Nov 2019 - Mar 2020

- Supported undergraduate students with C, C++, and Java coursework, assisting with debugging and core programming concepts in a collaborative learning environment.

PROJECTS

University Chatbot — LLM Fine-Tuning & RAG - LLaMA 2, DeepSeek, QLoRA, FAISS, Hugging Face

- Scraped SharePoint content into 600+ QA pairs; fine-tuned LLaMA 2 and DeepSeek with QLoRA and implemented FAISS-based RAG retrieval, improving answer relevance and reducing hallucinations.
- Benchmarked four model variants across retrieval and generation quality metrics; deployed the final model via Hugging Face Spaces.

AI-Powered Deepfake Voice Detection - CNN, Whisper, LoRA, Quantisation, FastAPI, Docker

- Engineered a hybrid CNN + Whisper pipeline for synthetic audio detection using MFCC features and transformer embeddings; applied LoRA fine-tuning and quantisation for efficient multilingual inference.
- Trained and evaluated on 220k+ real and synthetic audio samples; deployed as a FastAPI service containerised with Docker.

Credit Card Fraud Detection - Cloud AI, MLOps, CI/CD, GCP

- Built a modular, cloud-native fraud detection pipeline covering preprocessing, model registry, and real-time inference, with GDPR-compliant design and class-imbalance handling.
- Implemented CI/CD for automated testing and deployment, ensuring model reliability and scalability in a production-like environment.

SKILLS

Programming: Python, C#, .NET Core / ASP.NET, SQL, Java, C++, JavaScript, HTML/CSS, C

ML & AI: PyTorch, Hugging Face Transformers, Scikit-learn, LLM Fine-tuning (LoRA, QLoRA), NLP (ASR, NMT, RAG), TensorFlow, Whisper, Wav2Vec2, Model Quantisation

Cloud & DevOps: Docker, Kubernetes, CI/CD, GCP, AWS, Azure, Redis, RabbitMQ

Databases: PostgreSQL, MySQL, SQLite, Redis

Tools: Git, GitHub, Visual Studio, VSCode, Anaconda, Jupyter Notebook, Agile / Scrum, OPC, TCP/IP

Languages: English (Proficient), Turkish (Native)